Gesture Speak: Sign Gesture To Text & Voice Using CV & ML

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***Abstract*— Problems for disabled are being increased day-by- day .Its been too difficult to recognize the hand gestures by them with the one who are not familiar with them. This project explores the development of a system for converting sign language (ASL) to text and voice using computer vision techniques and Natural language processing (NLP). The process begins with the collection and annotation of video data, capturing individuals performing sign language. Frames are extracted from these live webcam, normalized, and subjected to feature extraction to identify key hand and body positions. Deep learning models which includes Convolutional Neural Networks and some models like Recurrent Neural Networks, are employed to recognize gestures and translate them into text. The system incorporates gesture recognition and language modeling to produce grammatically correct sentences. Tools such as OpenPose, TensorFlow, PyTorch, and OpenCV are utilized in the implementation. The project addresses challenges such as data availability in sign language, and real-time processing. It aims to improve communication tools for the disabled , hard-of-hearing, support sign language education, and improve accessibility in various applications. By combining NLP and computer vision, this research contributes to the development of efficient and accurate sign language translation systems.**

Keywords - Computer Vision , Natural Language Processing , OpenPose ,TensorFlow, PyTorch, OpenCV , ASL.

I. Introduction

Communication is fundamental to human interaction, yet individuals with hearing or speech impairments often face challenges in effectively expressing themselves, especially in environments where sign language is not widely understood. “Gesture Speak” is an innovative project aimed at bridging this communication gap by providing a solution that converts sign language gestures into text in real-time.

Gesture Speak is an innovative solution designed to cover the gap between communication and sign language users and nonsigners by converting sign language gestures into text in realtime using computer vision and machine learning, Gesture Speak ensures accurate and efficient translation of gestures to enable communication.

This means gesture recognition is such a technology that allows computers to read and then react to human gestures as interpreted by the computer for better communication between man and machine, which is extremely fluid in nature. It takes various forms, such as catching any physical movement of hand or even body posture or face, using sensors or cameras. The process involves three steps. Gesture acquisition, which first captures raw data using a device; preprocessing or data cleaning and normalization. Finally, key features of the gesture are extracted such as shapes, trajectories, or patterns of motion to classify them. Algorithms range from a basic template matching technique to machine learning techniques like deep and advanced models: Convolutional Neural Network-based approach for image recognition-based analysis and Recurrent Neural Networks to analyze dynamic gesture-based on the interpretation. The Fig. 1 describes the social model of disability which are faced by the disabled persons.

Fig.1 Problems of disabled persons in the society.

This document describe about introduction in Section-I of the project, the Literature Surveys in Section-II, Section-III consists of algorithm and it is followed by Section-IV which describes the methodology which has been done in our research study, Section-V says about the results obtained, Section-VI has conclusion and the future work of the research work and the final section has references which are used in this document.

II. Related Work

Hitesh sri sai Kaushik vezzu et al [1] developed his research on Hand sign recognition using image processing. His work mainly focuses on recognizing (American Sign Language) ASL . The user sign will be captured by camera and it is transalted .The draw back of his work is limited to ASL and changes to light conditions and background and also it is time taking process.

Saleh Aly et al [2] developed his research which is used for detection of the dynamic gestures. His work mainly focuses on the gestures in sign language which are dynamic in nature. The research deals about the analysis of this dynamic gesture. The two challenges in his work is hand shape and hand segmentation. ASD used the technology came from Hidden Markov Models. The disadvantages are it is less flexible and less user-friendly.

Muneer Al-Hammadi et al [3] developed his research on sign language detection using transfer learning and CNN . The aim of this project is to recognize the sign he gestures some certain metrics. This can be performed by three gesture datasets of colored videos. This transfer learning makes of a designing a large-labeled hand gesture dataset. This document has some drawbacks are signer independent mode and also the rate of recognition are less.

Wei Li et al [4] developed his research which is used for detection of traffic scenes. The vehicle technology need for understanding traffic scenes from detection of traffic sign recognition to vehicle detection of traffic scenes. The directions were provided from the traffic scenes through a set of tools incorporating natural language processing such as LSTM. The major draw back of this paper is that it needs a large trained data set.

Ilias Papastraits et al [5] developed his research which is used for detection of continuous sign language. The main objective of this study is to discuss about the challenges in sign language recognition which interpreting signs and determining the boundaries in potentially weakly annotated video. This has some feature extraction of textual information so that it is used for recognizing the visual observations. The methodology is based on cross-model learning. This study have some disadvantages like it requires more for preprocessing steps than normal.

Hassan et al [6] used gaussian type distribution for recognization gestures. They applied some operations for identifying shape of the hand and get its features and its direction analysis. They used hand slope to detect the direction of hand gesture so it helps to identify the gesture by using in gaussian type distribution. The draw back of his research work is it is complex for designing.

Tewodros Legesse Munea et al [7] developed his research which is used for identifying and detecting the human pose. It involves the identification of human posture key points to predict the actions done by the individual . The classification of the process actions depends on the behaviour marking of key points on the body.The field includes gaming, videos, human tracking etc. The drawbacks of the study is quite boring in nature ,it is complex to study and requires a lot of data for training.

Mohammed A.Bencherif et al [8] developed the Arabic sign language recognition using image processing techniques. Some hearing disabled may use Arabic sign language in communication with each other. This work is based on a system that makes use of three dimensional convolution neural networks , which provides a skeleton that captures gestures in every frame in a sequenced video and mapping them in Arabic script accordingly. The disadvantage of this work is that it is complex and not user-friendly and is restricted for Arabic sign language only.

Farman Shah Muhammad et al [9] done his research work on Pakistan sign language(PSL). Some communities belongs to dumb may use Pakistan sign language as their priority for communication. He mainly focus on the colour based hands . His work is to convert the PSL in to their native language. The main disadvantage of the work is it is costly and also it is not user friendly and his work is only about PSL.

Praveen Kumar et al [10] developed the research work which is used for detection of the sign language recognition. The main objective of this research work is for recognition of gestures by capturing and it converts to speech in its respective signs.They utilize the video frames that are noise free through skin color-based segmentation. Static and dynamic gesture differences and features were extracted. The disadvantages of his paper are it is less portable and not user friendly.

G. Csonka et al [11] developed a AI-based real-time hand gesture recognition system for robot control using transfer learning (ResNet) and a custom robot with Raspberry Pi and Arduino. It demonstrates promising applications in automation and smart homes, but its drawbacks include dependency on high-quality datasets, limited robustness under varying lighting/backgrounds, and potential hardware cost/complexity.

Oudah M et al [12] reviews computer vision–based hand gesture recognition techniques, discussing segmentation methods, classification algorithms, datasets, and applications. It highlights their merits and limitations, noting issues like accuracy variations, environmental sensitivity, and dataset dependence.

III. Algorithm

The study will consist of two main modules.The first module discussed about the creating the dataset and the second module discussed about the training and testing of data. The input is taken from the live camera and then the English words are converted in to speech or voice. For the module one an algorithm is designed for dataset creation using teachable machine and for module two algorithm is designed for training and testing of data and then a user interface is created.

Algorithm-1: Creation of a dataset

Input: Images from the web cam

Output: Collection of images with different gestures.

1. Setup
    1.1. Import required libraries
    1.2. Set parameters(offset, imgSize, folder path)
    1.3. Initialize video capture and HandDetector
    1.4. Initialize counter
2. Capture Video
    2.1. Start video capture loop
    2.2. Read a video frame
3. Detect Hand
    3.1. Use HandDetector to find hands
    3.2. If a hand is detected do
        Extract bounding box (x, y, w, h)
4. Crop and Resize
    4.1. Create a blank white canvas (imgWhite)
    4.2. Crop the hand region using bounding box + padding
    4.3. Adjust the aspect ratio
        If height > width do : Scale height to imgSize, adjust width proportionally, center horizontally

Else do : Scale width to imgSize, adjust height proportionally, center vertical

4.4. Display cropped image and processed canvas

5. Save Processed Images

5.1. Check for key press

If 's' key is pressed do

Save imgWhite to folder

Increment and print counter

5.2. Display the original frame

6. Exit Condition

If 'q' key is pressed do

Break the loop

7. Release Resources

Release video capture and close OpenCV windows

Algorithm-2: Training and Testing of dataset

Input: The images which are collected from the web cam during creation of dataset.

Output: A trained and tested model with a interface created for showing the text and voice also enabled.

1. Initialize

1.1. Import libraries

Numpy,cv2,math,pyttsx3,HandDetector

1.2. Load the trained gesture model

1.3. Define gesture labels and parameters(offset, imgSize, etc.)

1.4. Initialize pyttsx3 for Text-to-Speech

2. Open Webcam

2.1. Start video capture loop

2.2. Read a video frame

3. Detect Hand

3.1. Use HandDetector to find hands

3.2. If a hand is detected do

Extract bounding box dimensions(x, y, w, h)

4. Crop and Resize

4.1. Crop the hand image using bounding box + padding

4.2. Resize the cropped image based on aspect ratio

4.3. Center the resized image on a white canvas

5. Recognize Gesture

5.1. Pass the processed image (imgWhite) to the classifier

5.2. Get the gesture label using the predicted index

6. Display Text

6.1. Append label to detected_text

6.2. Display updated text on a separate window

7. Convert Text to Speech

7.1. Convert label to speech using pyttsx3

8. Visualize

8.1. Draw bounding box and display label on the main video feed.

8.2. Display the video frame

9. Exit Condition

9.1. If 'q' is pressed do

Break the loop

10. Release Resources

10.1. Release video capture and close OpenCV window

IV. Methodlogy

This research work has of two sections. By using these two modules we can identify the sign that the user is posing in front of camera. The first module focus on creation of dataset and the second module focus on testing and training the dataset and also a user interface is created to display the text and finally the voice is also enabled.

A. Module I

Set-up: First of all, the necessary libraries into the environment are imported those are required for proper work of the program. On the other hand, values such as offset, dimensions of the image and addresses of folders are set strictly to ensure proper functioning of the program.

Secondly, video capture is started in which module that is hand detector starts, helps in finding hands within video capture. Find hands after capture in every capture frame, the image of capture frame is processed line wise by hand detector and through this detector, region and location of hands in frames would be found.

Crop and resize: In order to display on the screen, blank canvas is created that acts a base for images, would be processed and will displayed on the screen. The area of the hand is cropped with precision using the bounding box method, which includes a padding feature for added precision and clarity. Aspect ratio adjustments are essential to ensure that the image size remains consistent across different displays and formats , it is also centered horizontally on the canvas for visual balance.

Finally, the processed version of the image along with the canvas is displayed prominently for viewing. Saving of image immediately when the user has typed the 's' key, it saves the processed image of the hand to a pre-set folder. Additionally, a counter is used by the system to track and update the quantity of images saved.

Resource Deallocation: All OpenCV windows opened during the process are neatly closed after the video capture resources have been appropriately deallocated.

According to Table I, the dataset comprises 3991 RGB images in the.jpg format, each of which was resized for training from its original size of 120×80×3. To ensure a fair assessment, 2443 of these were used for training and 1548 for testing. While optimization for latency, resource constraints, privacy, robustness, and usability guarantees the system's applicability in real-world scenarios, adaptation mechanisms enable customization, error correction, and learning over time.

TABLE I. IMAGE DETAILS IN DATASET

Label	Value
Image type	RGB
Image actual size	(120,80,3)
Image resized to	(300,200,3)
Image Format	*.jpg*
Training images	2443
Testing Images	1548
Total images	3991

B. MODULE II

This comprehensive framework effectively combines the core features of real-time text-to-speech (TTS) with the cutting-edge technology of hand gesture recognition, creating a very reliable solution intended for enhancing human-computer interaction and a variety of assistive applications. The well-designed framework consists of the following essential steps: Speech and hand movement signals should be recorded and preprocessed in order to extract features. Gestures should then be categorized, temporally aligned with speech, and used to drive system actions and spoken feedback.

Initialization: In this is first phase the necessary libraries are carefully imported, which include a wide range of tools specifically meant for video processing, hand detection, gesture classification, and the all-important TTS functionality, which in this case uses the library called pyttsx3 A pre-trained gesture recognition model is carefully loaded into the system to facilitate the process of recognition. To make sure that there is uniformity and consistency during the preprocessing phase, gesture labels along with important parameters like padding (which refers to the offset) and the target image size.

Video Capture Setup: in this step, a video feed is established from the webcam that is important for capturing video input thus allowing continuous frame acquisition in an ongoing loop.

Hand Detection: Every frame is processed by the use of a dedicated HandDetector module, which has been created to identify all the different hand regions present in the frame. The values represented by (x, y, w, h) for a successfully detected hand are obtained by carefully extracting the bounding box's dimensions, allowing for any further processing steps that may be required.

Image Preprocessing: It involves cropping the hand-containing portion of the image according to the bounding box's specified dimensions and adding padding to ensure that the entire hand is visible in the cropped area and no portion is missing.

Resized image: The resized image is then centered on the white blank canvas known as imgWhite. Because it ensures that the image will function with the gesture recognition model and be processed appropriately, this standardization process is essential.

Gesture Recognition: After preparation, the gesture classifier receives the preprocessed image which is imgWhite, for further processing. The model will return an index value that corresponds to a predefined gesture label, such as "stop" or "hello," after predicting the gesture shown in the image.

Text Display and Management: The recognized gesture label is append-only to a text display whose text will dynamically update at run time so that, to the viewer, a clearly visible display of how these gestures have been found may be produced. That text is shown in an optional separate window specifically aimed to improve user feedback

Text-to-Speech Conversion: The name assigned to the detected label goes through text-to-speach conversion, utilizing a module called pyttsx3 for converting it into heard speech, thereby making it offer immediate audio feedback to users on what it is going to do or what exactly is being performed on input devices. This particular framework serves to facilitate a seamless and efficient pipeline that is designed specifically for the purposes of detecting, recognizing, and converting various hand gestures into speech in real-time without any noticeable delay. It proves to be particularly ideal for a huge range of applications, such as aiding individuals who face challenges due to speech impairments, as well as being highly suitable for gesture-controlled devices and immersive user interfaces that require interactive capabilities.

Table II describes The model was trained on resized RGB images of size (300×200×3) with a batch size of 16 for up to 80 epochs, using early stopping on val_loss in min mode. Training took around 15 minutes with 20 validation steps per epoch to monitor performance. The model achieved 95.02% accuracy with a low loss of 0.16.

TABLE. II. RESULTS & METRICS OF TRAINING

Label	Value
Validation_steps	20
Early Stopping mode	Min
Early Stoppings monitor parameter	Val_loss
Training time	15 min (approx) (depends on resource allocation)
Batch Size	16
Classification Mode	Categorical
Target Size	(300,200,3)
Accuracy	95.02%
Loss	0.16
Epochs	80

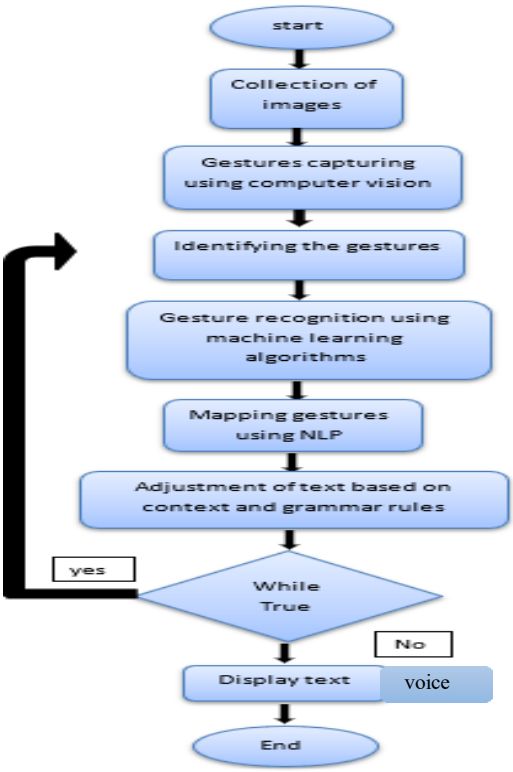

Fig. 2. Workflow of the proposed Gesture Speak system.

Fig.2. shows the framework captures and recognizes gestures using computer vision and machine learning, followed by mapping with NLP for grammatical accuracy. The output is displayed as text and optionally converted to voice for real-time communication.

V. Results

Our research was done by our proposed approach on test data. Our output is based on the input gesture given in front of the live camera. The output has the actual translation of the sign language based on given input and also the voice is generated according to the generated text. The estimated accuracy of this proposed work is above 95% .It satisfies the performances in real-time.

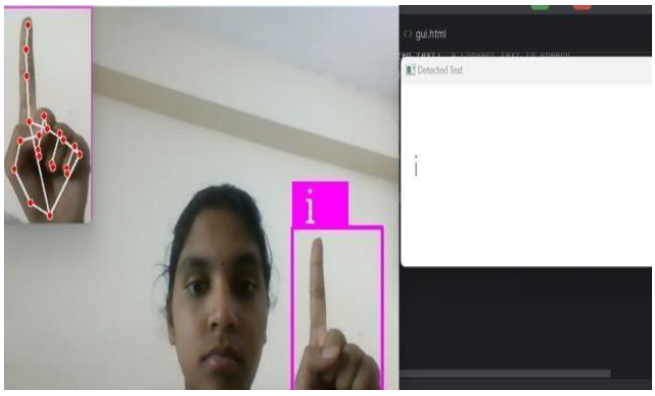

Fig 3: denotes displaying letter I.

This Fig 3 the system recognizes the gesture for the letter **"I"** and displays it on the interface.This demonstrates the model's ability to detect single-character gestures.It forms the base element for constructing complete sentences.

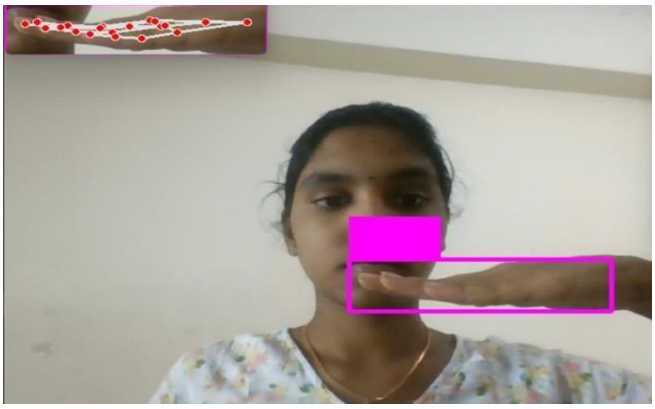

Fig. 4 denotes space.

This Fig 4 shows that the gesture for **"space"** is detected and processed by the system. This enables proper separation of words during sentence formation. So that it ensures clarity and grammatical correctness in the output.

Fig.5 identifies drink and sentence becomes I drink.

From Fig 5 The gesture **"drink"** is successfully recognized and displayed.
When combined with earlier detections, it generates **"I drink"**. This shows how sequential gestures form meaningful phrases.

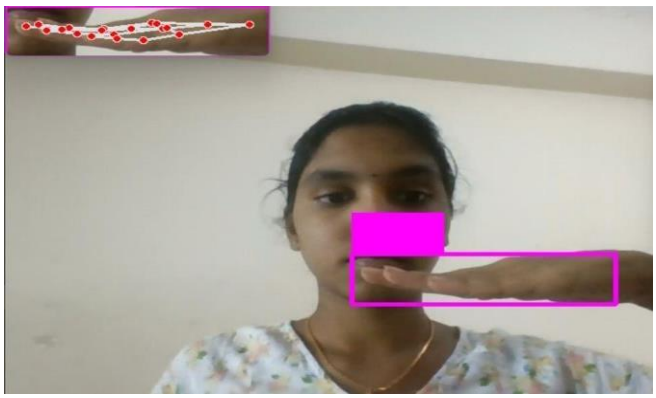

Fig.6 denotes space.

Another **"space"** gesture is identified by the system. It plays a key role in structuring sentences with multiple words.This confirms the consistency of the model in repeated detections.

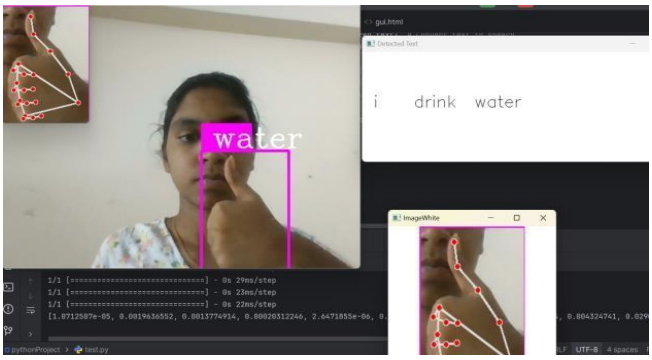

Fig.7 denotes the symbol water.

The figures 3,4,5,6,7 gives the sentence of **I Drink Water.** The above figures identify the respective gestures and give them as sentence and voice also produced.

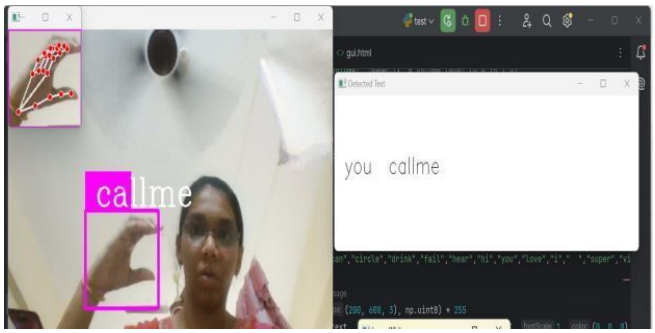

Fig.8 denotes you call me sentence.

You is identified first and then space is identified and then call me is identified .So the figure 8 gives the output as You call me. When the web cam starts it captures the symbols and the respective gestures are identified and the text is displayed on the interface along with this voice also will be heard.

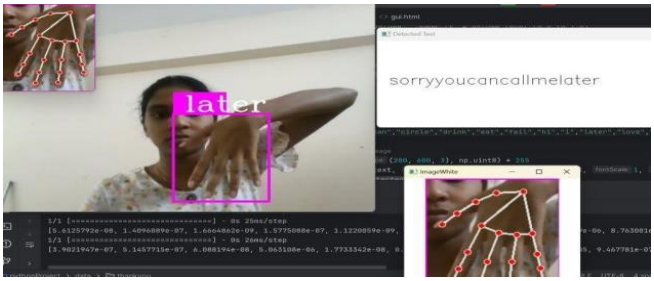

Fig.9 denotes sorry you can call me later sentence.

This figure denotes Sorry you can call me later.

VI. CONCLUSION & FUTIRE WORK

The proposed work makes use of Convolutional Neural Networks (CNNs) to recognize and classify Sign Language gestures with high accuracy. It not only recognizes gestures but also converts the detected gesture into text and voice through the use of text-to-speech technology, providing better accessibility for users with different requirements. The system achieved over 95% accuracy and works in real time.
In the future, we plan to extend this system to support more sign languages, improve its performance in different lighting and background conditions, and make it available as a mobile application. We also aim to enhance the system's ability to recognize continuous gestures and improve user experience by creating a more interactive and user-friendly interface. Furthermore, we plan to expand the dataset by including larger vocabularies of static and dynamic signs collected under diverse conditions to enhance scalability and robustness. In addition, we intend to conduct user studies with hearing and speech-impaired individuals to validate the system's practical usability and gather feedback for improving accessibility and user acceptance.

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